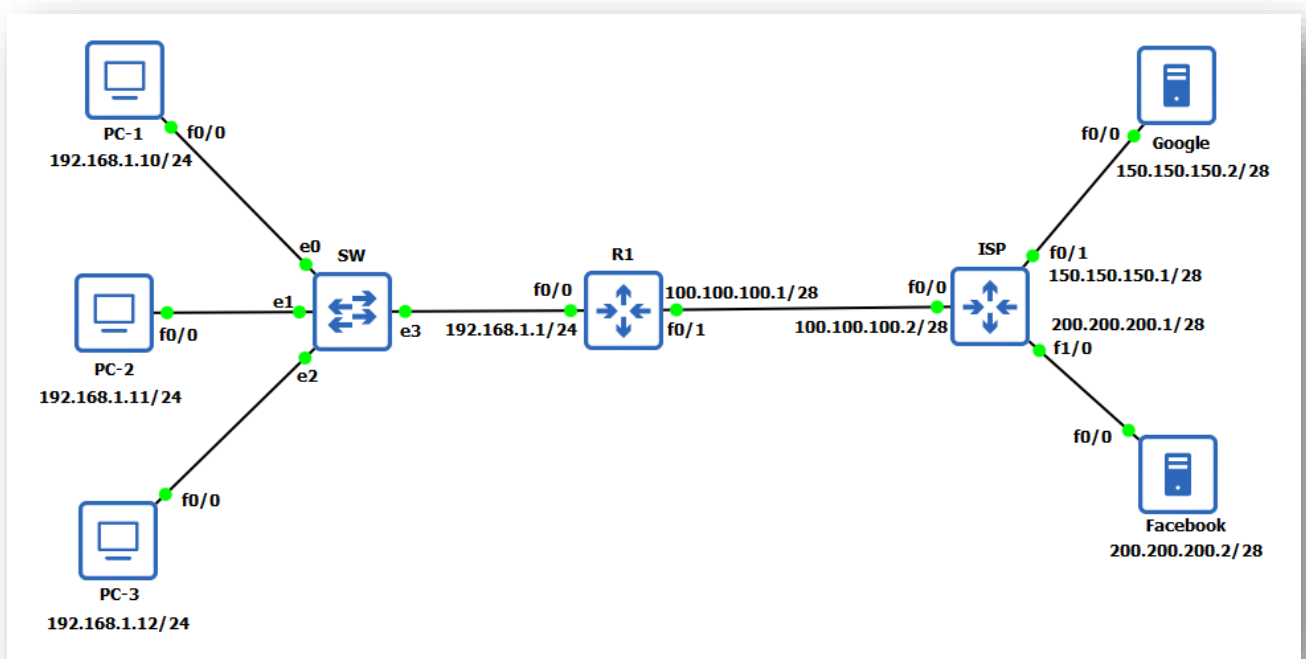


Dynamic NAT



Task

1. Configure routers R1 & ISP with IP address as shown in topology and configure enable password as **ccna**. (**Refer Lab-19 for task-1 to task-4**)
2. Configure routers as Host (PC)
3. Configure routers as Host (Servers) and enable HTTP service on it.
4. Configure default routing on R1.
5. Configure Dynamic NAT with pool of 2 Public IP for 192.168.1.0/24 network.

Task-5: Configure Dynamic NAT.

```

R1#config t
R1(config)#access-list 10 permit 192.168.1.0 0.0.0.255
R1(config)#ip nat pool Public-IP 100.100.100.3 100.100.100.4 prefix-length 28
R1(config)#ip nat inside source list 10 pool Public-IP
R1(config)#int fa0/0
R1(config-if)#ip nat inside
R1(config-if)#exit
R1(config)#int fa0/1
R1(config-if)#ip nat outside
R1(config-if)#exit
R1(config)#exit
R1#

```

✓ Verification & Testing:

R1#sh ip nat translations

Ping Google and Facebook server starting from PC-1, PC-2 and PC-3

PC-1#ping 150.150.150.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 150.150.150.2, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 52/59/72 ms

PC-1#

PC-1#ping 200.200.200.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 200.200.200.2, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 40/56/64 ms

PC-1#

PC-1#telnet 150.150.150.2 80

Trying 150.150.150.2, 80 ... Open

PC-1#

PC-1#telnet 200.200.200.2 80

Trying 200.200.200.2, 80 ... Open

PC-1#

PC-2#ping 150.150.150.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 150.150.150.2, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 56/62/80 ms

PC-2#

PC-2#ping 200.200.200.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 200.200.200.2, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 40/54/60 ms

PC-2#

PC-2#telnet 150.150.150.2 80

Trying 150.150.150.2, 80 ... Open

PC-2#

PC-2#telnet 200.200.200.2 80

Trying 200.200.200.2, 80 ... Open

PC-2#

PC-3#ping 150.150.150.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 150.150.150.2, timeout is 2 seconds:

UUUUU

Success rate is 0 percent (0/5)

PC-3#

PC-3#ping 200.200.200.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 200.200.200.2, timeout is 2 seconds:

UUUUU

Success rate is 0 percent (0/5)

PC-3#

PC-3#telnet 150.150.150.2 80

Trying 150.150.150.2, 80 ...

% Destination unreachable; gateway or host down

PC-3#

PC-3#telnet 200.200.200.2 80

Trying 200.200.200.2, 80 ...

% Destination unreachable; gateway or host down

PC-3#

R1#sh ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
icmp	100.100.100.3:0	192.168.1.10:0	150.150.150.2:0	150.150.150.2:0
icmp	100.100.100.3:1	192.168.1.10:1	200.200.200.2:1	200.200.200.2:1
tcp	100.100.100.3:17700	192.168.1.10:17700	150.150.150.2:80	150.150.150.2:80
tcp	100.100.100.3:29975	192.168.1.10:29975	200.200.200.2:80	200.200.200.2:80
icmp	100.100.100.4:2	192.168.1.11:2	150.150.150.2:2	150.150.150.2:2
icmp	100.100.100.4:3	192.168.1.11:3	200.200.200.2:3	200.200.200.2:3
tcp	100.100.100.4:17132	192.168.1.11:17132	200.200.200.2:80	200.200.200.2:80
tcp	100.100.100.4:23099	192.168.1.11:23099	150.150.150.2:80	150.150.150.2:80
tcp	100.100.100.4:49429	192.168.1.11:49429	200.200.200.2:80	200.200.200.2:80
tcp	100.100.100.4:60057	192.168.1.11:60057	150.150.150.2:80	150.150.150.2:80
---	100.100.100.4	192.168.1.11	---	---

R1#

R1#sh ip nat statistics

Total active translations: 8 (0 static, 8 dynamic; 6 extended)

Peak translations: 10, occurred 00:02:39 ago

Outside interfaces:

FastEthernet0/1

Inside interfaces:

FastEthernet0/0

Hits: 200 Misses: 0

CEF Translated packets: 200, CEF Punted packets: 10

Expired translations: 7

Dynamic mappings:

-- Inside Source

[Id: 1] access-list 10 pool Public-IP refcount 8

pool Public-IP: netmask 255.255.255.240

start 100.100.100.3 end 100.100.100.4

type generic, total addresses 2, allocated 2 (100%), misses 10

Total doors: 0

Appl doors: 0

Normal doors: 0

Queued Packets: 0

R1#

Enable icmp debugging on PC-1 & PC-2. Ping NATed IP from Google and Facebook server.

PC-1#debug ip icmp

ICMP packet debugging is on

PC-1#

PC-2#debug ip icmp

ICMP packet debugging is on

PC-2#

Google#ping 100.100.100.3

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 100.100.100.3, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 44/55/68 ms

Google#

Google#ping 100.100.100.4

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 100.100.100.4, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 48/57/60 ms

Google#

Facebook#ping 100.100.100.3

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 100.100.100.3, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 40/56/64 ms

Facebook#

Facebook#ping 100.100.100.4

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 100.100.100.4, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 44/56/60 ms

Facebook#

PC-1#

*Nov 14 19:34:35.883: ICMP: echo reply sent, src 192.168.1.10, dst 150.150.150.2, topology BASE, dscp 0 topoid 0

*Nov 14 19:34:35.939: ICMP: echo reply sent, src 192.168.1.10, dst 150.150.150.2, topology BASE, dscp 0 topoid 0

*Nov 14 19:34:35.999: ICMP: echo reply sent, src 192.168.1.10, dst 150.150.150.2, topology BASE, dscp 0 topoid 0

*Nov 14 19:34:36.059: ICMP: echo reply sent, src 192.168.1.10, dst 150.150.150.2, topology BASE, dscp 0 topoid 0

*Nov 14 19:34:36.123: ICMP: echo reply sent, src 192.168.1.10, dst 150.150.150.2, topology BASE, dscp 0 topoid 0

PC-1#

*Nov 14 19:34:44.555: ICMP: echo reply sent, src 192.168.1.10, dst 200.200.200.2, topology BASE, dscp 0 topoid 0

*Nov 14 19:34:44.611: ICMP: echo reply sent, src 192.168.1.10, dst 200.200.200.2, topology BASE, dscp 0 topoid 0

*Nov 14 19:34:44.671: ICMP: echo reply sent, src 192.168.1.10, dst 200.200.200.2, topology BASE, dscp 0 topoid 0

*Nov 14 19:34:44.731: ICMP: echo reply sent, src 192.168.1.10, dst 200.200.200.2, topology BASE, dscp 0 topoid 0

*Nov 14 19:34:44.795: ICMP: echo reply sent, src 192.168.1.10, dst 200.200.200.2, topology BASE, dscp 0 topoid 0

PC-1#

PC-2#

*Nov 14 19:34:38.275: ICMP: echo reply sent, src 192.168.1.11, dst 150.150.150.2, topology BASE, dscp 0 topoid 0

*Nov 14 19:34:38.335: ICMP: echo reply sent, src 192.168.1.11, dst 150.150.150.2, topology BASE, dscp 0 topoid 0

*Nov 14 19:34:38.395: ICMP: echo reply sent, src 192.168.1.11, dst 150.150.150.2, topology BASE, dscp 0 topoid 0

*Nov 14 19:34:38.459: ICMP: echo reply sent, src 192.168.1.11, dst 150.150.150.2, topology BASE, dscp 0 topoid 0

*Nov 14 19:34:38.519: ICMP: echo reply sent, src 192.168.1.11, dst 150.150.150.2, topology BASE, dscp 0 topoid 0

PC-2#

*Nov 14 19:34:46.331: ICMP: echo reply sent, src 192.168.1.11, dst 200.200.200.2, topology BASE, dscp 0 topoid 0

*Nov 14 19:34:46.375: ICMP: echo reply sent, src 192.168.1.11, dst 200.200.200.2, topology BASE, dscp 0 topoid 0

*Nov 14 19:34:46.439: ICMP: echo reply sent, src 192.168.1.11, dst 200.200.200.2, topology BASE, dscp 0 topoid 0

*Nov 14 19:34:46.503: ICMP: echo reply sent, src 192.168.1.11, dst 200.200.200.2, topology
BASE, dscp 0 topoid 0
*Nov 14 19:34:46.563: ICMP: echo reply sent, src 192.168.1.11, dst 200.200.200.2, topology
BASE, dscp 0 topoid 0
PC-2#

R1#sh arp

Protocol	Address	Age (min)	Hardware Addr	Type	Interface
Internet	100.100.100.1	-	ca07.27a0.0006	ARPA	FastEthernet0/1
Internet	100.100.100.2	10	ca04.2360.0008	ARPA	FastEthernet0/1
Internet	100.100.100.3	-	ca07.27a0.0006	ARPA	FastEthernet0/1
Internet	100.100.100.4	-	ca07.27a0.0006	ARPA	FastEthernet0/1
Internet	192.168.1.1	-	ca07.27a0.0008	ARPA	FastEthernet0/0
Internet	192.168.1.10	11	ca01.1f60.0008	ARPA	FastEthernet0/0
Internet	192.168.1.11	10	ca02.1be8.0008	ARPA	FastEthernet0/0
Internet	192.168.1.12	7	ca03.287c.0008	ARPA	FastEthernet0/0

R1#

Now clear the NAT table and ping NATed IP from Google and Facebook server.

R1#clear ip nat translation *

R1#sh ip nat translations

R1#

Google#ping 100.100.100.3

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 100.100.100.3, timeout is 2 seconds:

.....

Success rate is 0 percent (0/5)

Google#

Google#ping 100.100.100.4

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 100.100.100.4, timeout is 2 seconds:

.....

Success rate is 0 percent (0/5)

Google#

Facebook#ping 100.100.100.3

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 100.100.100.3, timeout is 2 seconds:

.....

Success rate is 0 percent (0/5)

Facebook#

Facebook#ping 100.100.100.4

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 100.100.100.4, timeout is 2 seconds:

.....

Success rate is 0 percent (0/5)

Facebook#

Note: We can initiate traffic from outside on NATed public IP only when we have NAT translation entries available.

To Remove Dynamic NAT

```
R1#config t
R1(config)#no ip nat inside source list 10 pool Public-IP
Dynamic mapping in use, do you want to delete all entries? [no]: yes
R1(config)#no access-list 10 permit 192.168.1.0 0.0.0.255
R1(config)#no ip nat pool Public-IP 100.100.100.3 100.100.100.4 prefix-length 28
R1(config)#int fa0/0
R1(config-if)#no ip nat inside
R1(config-if)#exit
R1(config)#int fa0/1
R1(config-if)#no ip nat outside
R1(config-if)#exit
R1(config)#exit
R1#
```

Important Commands:

sh ip nat translations
sh ip nat statistics
sh arp

Ratan